



BAULKHAM HILLS HIGH SCHOOL

2019

HSC Assessment Task 1

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 50 minutes
- Write using non-erasable black or blue pen.
- Calculators approved by NESA may be used.
- Marks may be deducted for careless or badly arranged work.
- Relevant formulae from the NESA Reference Sheet are provided on page 2.
- Answer all questions in the space provided.
- For questions in Section II, show relevant mathematical reasoning and/or calculations.
- If extra paper is used, it must be clearly labelled with your name and the question being answered, and stapled inside your answer booklet.

Total marks – 45

Exam consists of 12 pages.

Section I: Pages 3-4

5 marks

Attempt Questions 1-5

Section II: Pages 4-12

40 marks

Attempt Questions 6-15.

Complex Numbers

$$z = a + ib = r(\cos \theta + i \sin \theta) \\ = re^{i\theta}$$

$$\left[r(\cos \theta + i \sin \theta)\right]^n = r^n(\cos n\theta + i \sin n\theta) \\ = r^n e^{in\theta}$$

Mathematics Extension 2

Section I – Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$

(A) 2
A ☐

(B) 6
B ☒

(C) 8
C ☐

(D) 9
D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒

B ☒

C ☐

D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒

B ☒

correct

C ☐

D ☐

**Start
Here** →

1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

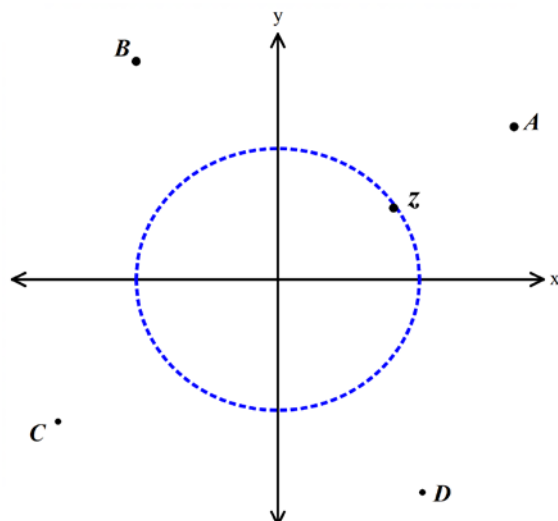
3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

5. A ☐ B ☐ C ☐ D ☐

Section I: 5 marks Attempt Questions 1-5 Use the multiple choice answer sheet on the previous page.		Marks
1	Let $z = 2 - 3i$ and $w = 3 + 2i$. What is the value of $z - \bar{w}$? (A) $5 - i$ (B) $-1 - 5i$ (C) $-1 - i$ (D) 0	1
2	If z is a complex number, which of the following is equivalent to $z\bar{z} - z ^2$? (A) 1 (B) 0 (C) $(z + \bar{z})(z - \bar{z})$ (D) $(z + iz)(z - iz)$	1
3	Let $z = a + bi$, where a and b are non-zero real numbers. If $z + \frac{1}{z}$ is purely real, which of the following statements must be true? (A) $a = b$ (B) $a = -b$ (C) $z = 1$ (D) $z^2 = 1$	1
4	When the complex number $-1 + 5i$ is written in the form $\sqrt{26}e^{i\theta}$, which of the following is the approximate value of θ ? (A) -1.77 (B) -1.37 (C) 1.37 (D) 1.77	1

5 The complex number z is shown on the Argand diagram below.



Which of the following best represents $-2iz$?

(A) Point A

(B) Point B

(C) Point C

(D) Point D

End of Section I

Section II: 40 marks

Attempt Questions 6-15 in the space provided.

Question 6 (5 marks)

Let $z = 1 - 2i$ and $w = 3 + i$. Find in the form $x + iy$ (where x and y are real numbers):

(i) zw

1

(ii) $z^2 - w^2$

2

(iii) $\overline{\left(\frac{10}{z}\right)}$

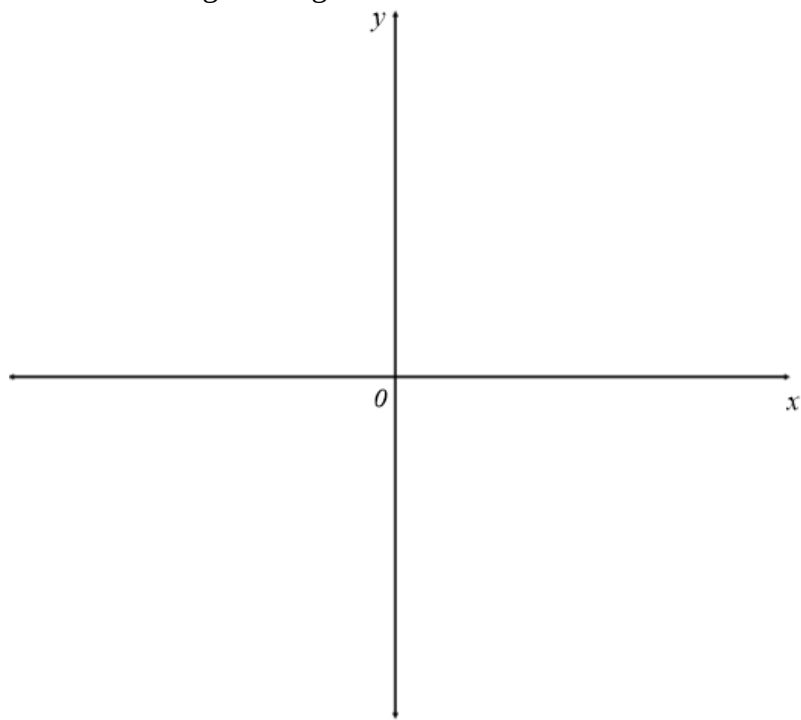
2

Question 7 (3 marks)

If $|z - 2| = |z - i|$:

(i) Sketch the locus of z on an Argand diagram.

2



(ii) Find the equation of the locus of z in Cartesian form.

1

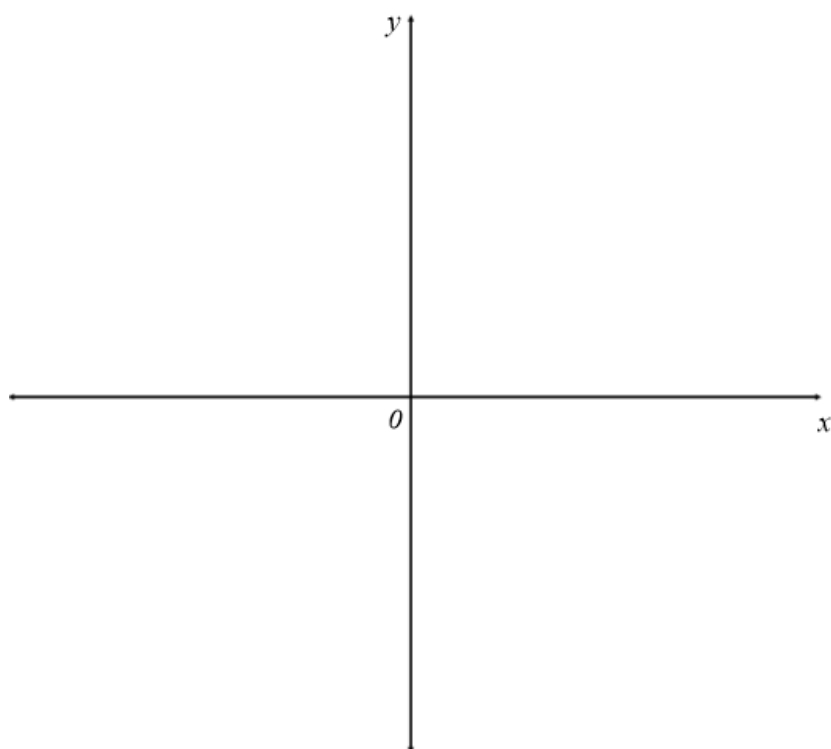
[illegible]

Question 8 (3 marks)

Given that $|z + 2 - 3i| = 2$:

- (i) Draw the locus of z on the Argand diagram.

2



- (ii) Find the maximum value of $|z|$.

1

Question 9 (5 marks)

Given $z = \sqrt{3} - i$

(i) Express z in modulus-argument form.

2

(ii) Hence or otherwise, evaluate z^{10} , giving your answer in Cartesian form.

2

(iii) What is the value of the least positive integer n for which z^n is purely imaginary?

1

Question 10 (4 marks)

- (i) Find the complex numbers z such that $z^2 = 5 + 12i$. Answer in Cartesian form. 2

- (ii) Hence or otherwise, solve the equation $z^2 + 3z + 1 - 3i = 0$. 2

Question 11 (3 marks)

- (i) Expand $(\cos\theta + i\sin\theta)^3$. 1

- (ii) Hence express $\sin 3\theta$ in terms of $\sin \theta$. 2

Question 12 (5 marks)

The polynomial $P(z) = 3z^4 - 4z^3 - 6z^2 + 20z - 16$ has a zero at $z = 1 + i$.

- (i) Find a quadratic factor of $P(z)$.

2

- (ii) Express $P(z)$ as a product of linear factors.

3

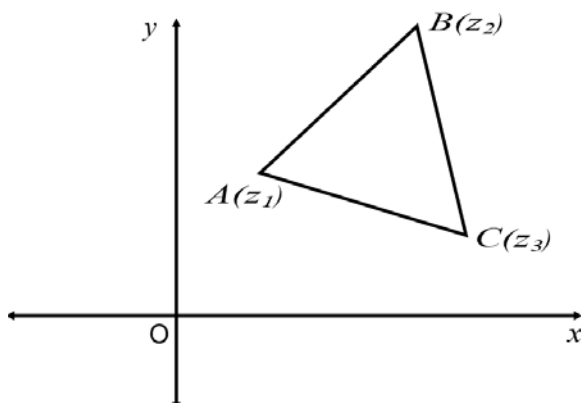
Question 13 (2 marks)

Simplify $(e^{i\theta} - e^{-i\theta})^4$, expressing your answer as a trigonometric function.

2

Question 14 (4 marks)

In the diagram below, $\triangle ABC$ is an equilateral triangle with $\overrightarrow{OA} = z_1$, $\overrightarrow{OB} = z_2$, $\overrightarrow{OC} = z_3$ and $\arg(z_2 - z_1) = \alpha$:



- (i) Find $\arg(z_2 - z_3)$ in terms of α .

1

- (ii) Find the modulus and argument of $\frac{z_2 - z_1}{z_3 - z_2}$.

3

Question 15 (6 marks)

Given $P(z) = z^5 - 1$ and $z \in \mathbb{C}$:

- (i) Find the roots of $P(z) = 0$.

2

- (ii) Express $P(z)$ as a product of real linear and real irreducible quadratic factors.

2

- (iii) Using the identity:

$$z^5 - 1 = (z - 1)(z^4 + z^3 + z^2 + z + 1)$$

Find the value of :

$$(1 + \cos \frac{2\pi}{5})(1 + \cos \frac{4\pi}{5})$$

2

END OF EXAMINATION

Section I:
5 marks
Attempt Questions 1-5

Use the multiple choice answer sheet on the previous page.

- 1 Let $z = 2 - 3i$ and $w = 3 + 2i$.
What is the value of $z - w$? 1

- (A) $5 - i$
(B) $-1 - 5i$
(C) $-1 - i$
(D) 0

$$2 - 3i - (3 + 2i) \\ = -1 - i$$

- 2 If z is a complex number, which of the following is equivalent to $z\bar{z} - |z|^2$? 1

- (A) 1
(B) 0
(C) $(z + \bar{z})(z - \bar{z})$
(D) $(z + iz)(z - iz)$

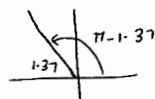
$$z\bar{z} = |z|^2 \therefore 0$$

- 3 Let $z = a + bi$, where a and b are non-zero real numbers. If $z + \frac{1}{z}$ is purely real, which of the following statements must be true? 1

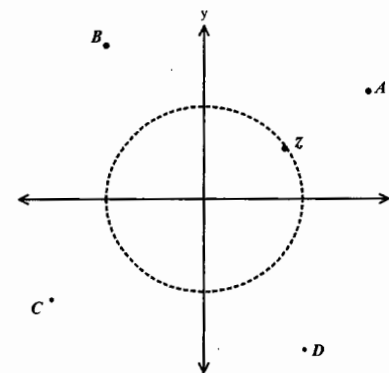
- (A) $a = b$
(B) $a = -b$
(C) $|z| = 1$
(D) $z^2 = 1$

- 4 When the complex number $-1 + 5i$ is written in the form $\sqrt{26}e^{i\theta}$, which of the following is the approximate value of θ ? 1

- (A) -1.77
(B) -1.37
(C) 1.37
(D) 1.77



- 5 The complex number z is shown on the Argand diagram below.



Which of the following best represents $-2iz$?

- (A) Point A (B) Point B (C) Point C (D) Point D

End of Section I

Section II: 40 marks
Attempt Questions 6-15 in the space provided.

Question 6 (5 marks)

Let $z = 1 - 2i$ and $w = 3 + i$. Find in the form $x + iy$ (where x and y are real numbers):

- (i) zw 1

$$= (1 - 2i)(3 + i) = 3 + 2 - 6i + i \\ = 5 - 5i \quad \text{1 - correct answer}$$

- (ii) $z^2 - w^2$ 2

$$= (1 - 2i - 3 - i)(1 - 2i + 3 + i) = -8 + 2i - 12i - 3 \\ = (-2 - 3i)(4 - i) = -11 + 10i \quad \begin{array}{l} \text{1 - partially correct} \\ \text{working} \\ \text{2 - correct answer} \end{array}$$

- (iii) $\left(\frac{10}{z}\right)$ 2

$$= \frac{10}{z} = \frac{10}{1 - 2i} \times \frac{1 - 2i}{1 - 2i} \\ = \frac{10(1 - 2i)}{5} = 2(1 - 2i) \quad \text{or } 2 - 4i$$

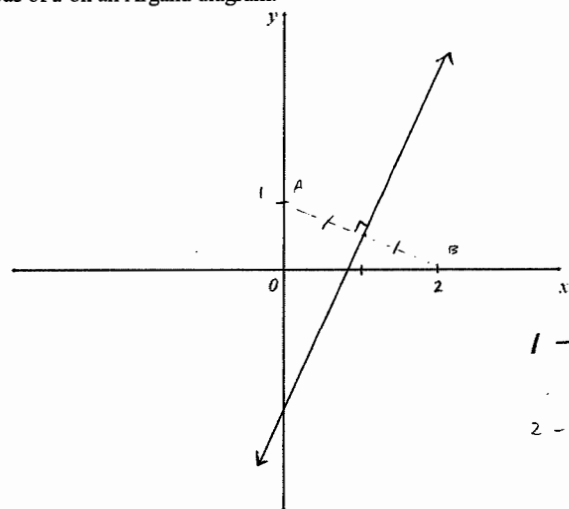
1 - attempt to rationalise a denominator
or express as $\frac{10}{1 - 2i}$
2 - correct solution

Question 7 (3 marks)

If $|z - 2| = |z - i|$:

(i) Sketch the locus of z on an Argand diagram.

2



1 - a straight line

2 - correct solution

(ii) Find the equation of the locus of z in Cartesian form.

1

Perpendicular bisector of $A(0, 1)$ and $B(2, 0)$:

$$m = 2, \text{ midpt} = \left(\frac{1}{2}, \frac{1}{2}\right)$$

$$y - \frac{1}{2} = 2\left(x - \frac{1}{2}\right)$$

$$y - \frac{1}{2} = 2x - 1$$

$$4x - 2y - 3 = 0 \quad \text{or} \quad y = 2x - \frac{3}{2}$$

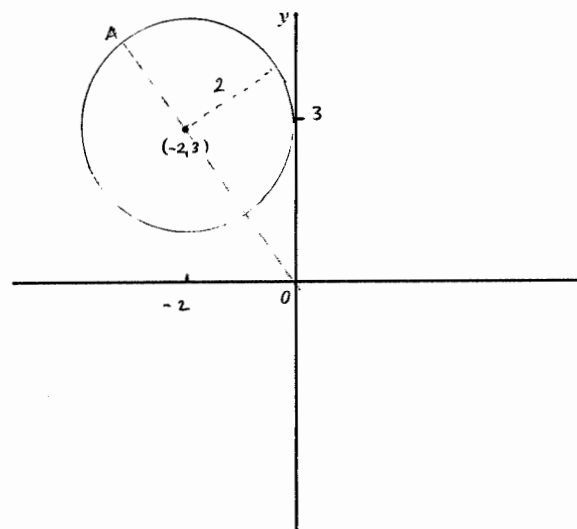
1 - correct eqn

Question 8 (3 marks)

Given that $|z + 2 - 3i| = 2$:

(i) Draw the locus of z on the Argand diagram.

2



1 - circle with correct centre or radius

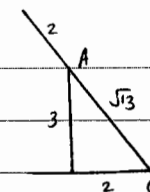
2 - correct circle

(ii) Find the maximum value of $|z|$.

1

Maximum value of $|z|$ occurs at A

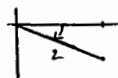
$$\text{Max. value of } |z| = \sqrt{13} + 2$$



1 - correct solution

Question 9 (5 marks)

Given $z = \sqrt{3} - i$



(i) Express z in modulus-argument form.

$$2\left(\cos -\frac{\pi}{6} + i \sin -\frac{\pi}{6}\right)$$

1 - correct mod or arg

$$\text{or } 2 \operatorname{cis} -\frac{\pi}{6}$$

2 - both correct

(must be principal arg)

(ii) Hence or otherwise, evaluate z^{10} , giving your answer in Cartesian form.

$$z^{10} = 2^{10} \operatorname{cis} \left(10 \times -\frac{\pi}{6}\right)$$

$$= 1024 \operatorname{cis} -\frac{5\pi}{3}$$

1 - attempts to apply De Moivre's theorem

$$= 1024 \operatorname{cis} \frac{\pi}{3}$$

2 - correct soln

$$= 1024 \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$$

$$= 512 + 512i\sqrt{3}$$

(iii) What is the value of the least positive integer for which z^n is purely imaginary?

1

If purely imaginary, $n \times \frac{\pi}{6}$ is an odd multiple of $\frac{\pi}{2}$.

$n=3$ is the first such value.

1 - correct answer.

Question 10 (4 marks)

(i) Find the complex numbers z such that $z^2 = 5 + 12i$. Answer in Cartesian form.

2

Let $z = x + iy$, $(x + iy)^2 = 5 + 12i$	$x^2 - 5x^2 - 36 = 0$
$x^2 - y^2 = 5 \quad (1)$	$(x^2 - 9)(x^2 + 4) = 0$
$2xy = 12$	$x = \pm 3, y = \pm 2$
$xy = 6 \quad (2)$	$z = \pm(3 \pm 2i)$
$x^2 - \left(\frac{6}{x}\right)^2 = 5$	1 - obtains a correct pair of simultaneous equations and attempts to solve

(ii) Hence or otherwise, solve the equation $z^2 + 3z + 1 - 3i = 0$.

2

$$\Delta = b^2 - 4ac = 3^2 - 4(1 - 3i) = 5 + 12i$$

$$z = \frac{-3 \pm \sqrt{5 + 12i}}{2}$$

2

$$= \frac{-3 \pm (3 + 2i)}{2}$$

2

$$= i, -3 - i$$

1 - applies quadratic formula with correct Δ .

2 - correct solutions

Question 11 (3 marks)

(i) Expand $(\cos \theta + i \sin \theta)^3$.

1

$$\begin{aligned} & \cos^3 \theta + 3 \cos^2 \theta \cdot i \sin \theta + 3 \cos \theta \cdot i^2 \sin^2 \theta + i^3 \sin^3 \theta \\ &= \cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta \end{aligned}$$

1 - either of above lines.

(ii) Hence express $\sin 3\theta$ in terms of $\sin \theta$.

2

$$(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta$$

$$\sin 3\theta = 3 \cos^2 \theta \sin \theta - \sin^3 \theta$$

$$= 3(1 - \sin^2 \theta) \sin \theta - \sin^3 \theta$$

1 - equates $\sin 3\theta$

$$= 3 \sin \theta - 4 \sin^3 \theta$$

with imaginary part of i .

2 - correct solution

Question 12 (5 marks)

The polynomial $P(z) = 3z^4 - 4z^3 - 6z^2 + 20z - 16$ has a zero at $z = 1 + i$.

- (i) Find a quadratic factor of $P(z)$.

2

Since coefficients are real, $1-i$ is also a root.

$$\alpha = 1+i, \beta = 1-i$$

1 - identifies $1-i$ is a root

$$\left. \begin{aligned} \alpha + \beta &= 2 \\ \alpha \beta &= 2 \end{aligned} \right\} \therefore z^2 - 2z + 2$$

2 - correct factor

- (ii) Express $P(z)$ as a product of linear factors.

3

$$P(z) = (z^2 - 2z + 2)(3z^2 + kz - 8)$$

Comparing terms in z :

$$16z + 2kz = 20z$$

$$k = 2$$

$$\therefore P(z) = (z^2 - 2z + 2)(3z^2 + 2z - 8)$$

$$= (z - 1 - i)(z - 1 + i)(3z - 4)(z + 2)$$

- obtains $(z - 1 - i)(z - 1 + i)$
- 1 - obtains $(3z^2 + 2z - 8)(z^2 - 2z + 2)$
- 2 - $(z - 1 - i)(z - 1 + i)(z - \frac{4}{3})(z + 2)$, units lead coefficient
- $(z - 1 - i)(z - 1 + i)$ and incorrect factors for the quadratic
- 3 - correct soln.

Question 13 (2 marks)

Simplify $(e^{i\theta} - e^{-i\theta})^4$, expressing your answer as a trigonometric function.

2

$$= (\cos \theta + i \sin \theta - (\cos(-\theta) + i \sin(-\theta)))^4$$

$$= (\cos \theta + i \sin \theta - (\cos \theta - i \sin \theta))^4$$

$$= (2i \sin \theta)^4$$

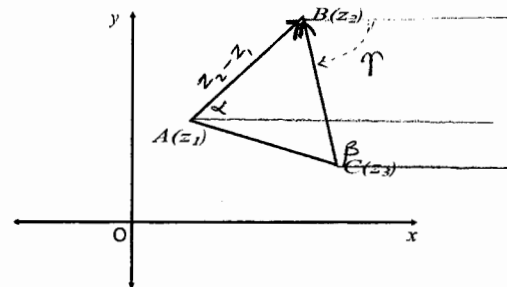
1 - significant progress

2 - correct solution

$$= 16 \sin^4 \theta$$

Question 14 (4 marks)

In the diagram below, $\triangle ABC$ is an equilateral triangle with $\overrightarrow{OA} = z_1$, $\overrightarrow{OB} = z_2$, $\overrightarrow{OC} = z_3$ and $\arg(z_2 - z_1) = \alpha$:



- (i) Find $\arg(z_2 - z_3)$ in terms of α .

1

$$\text{Let } \beta = \arg(z_2 - z_3)$$

$$(\beta + \frac{\pi}{3}) + (\frac{\pi}{3} - \alpha) = \pi \quad (\text{interior } \angle \text{ s, } // \text{ lines})$$

$$\beta + \frac{2\pi}{3} - \alpha = \pi$$

$$\therefore \beta = \arg(z_2 - z_3) = \alpha + \frac{\pi}{3}$$

1 - correct solution

- (ii) Find the modulus and argument of $\frac{z_2 - z_1}{z_3 - z_2}$.

3

$$\text{Modulus} = 1 \quad (\text{since } \triangle \text{ is equilateral})$$

$$\arg\left(\frac{z_2 - z_1}{z_3 - z_2}\right) = \arg(z_2 - z_1) - \arg(z_3 - z_2)$$

$$= \alpha - (\pi - \beta)$$

$$= \alpha - \pi + \alpha + \frac{\pi}{3}$$

$$= 2\alpha - \frac{2\pi}{3}$$

(1) - correct modulus

Then

(1) significant progress for arg.
(2) correct soln

Question 12 (5 marks)

The polynomial $P(z) = 3z^4 - 4z^3 - 6z^2 + 20z - 16$ has a zero at $z = 1 + i$.

(i) Find a quadratic factor of $P(z)$.

2

Since coefficients are real, $1-i$ is also a root.

$$\alpha = 1+i, \beta = 1-i$$

1 - identifies $1-i$ is a root

$$\alpha + \beta = 2$$

$$\alpha \beta = 2$$

$$\therefore z^2 - 2z + 2$$

2 - correct factor

(ii) Express $P(z)$ as a product of linear factors.

3

$$P(z) = (z^2 - 2z + 2)(3z^2 + kz - 8)$$

Comparing terms in z :

$$16z + 2kz = 20z$$

$$k = 2$$

$$\therefore P(z) = (z^2 - 2z + 2)(3z^2 + 2z - 8)$$

$$= (z - 1 - i)(z - 1 + i)(3z - 4)(z + 2)$$

1 - obtains $(z - 1 - i)(z - 1 + i)$

2 - obtains $(3z^2 + 2z - 8)(z^2 - 2z + 2)$

3 - $(z - 1 - i)(z - 1 + i)(z - \frac{4}{3})(z + 2)$, omits lead coefficient

4 - $(z - 1 - i)(z - 1 + i)$ and incorrect factors for the quadratic

5 - correct soln.

Question 13 (2 marks)

Simplify $(e^{i\theta} - e^{-i\theta})^4$, expressing your answer as a trigonometric function.

2

$$= (\cos \theta + i \sin \theta - (\cos(-\theta) + i \sin(-\theta)))^4$$

$$= (\cos \theta + i \sin \theta - (\cos \theta - i \sin \theta))^4$$

$$= (2i \sin \theta)^4$$

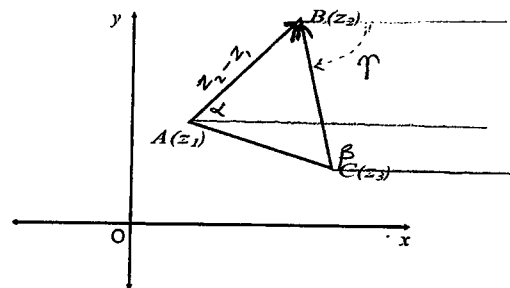
1 - significant progress

2 - correct solution

$$= 16 \sin^4 \theta$$

Question 14 (4 marks)

In the diagram below, $\triangle ABC$ is an equilateral triangle with $\overrightarrow{OA} = z_1$, $\overrightarrow{OB} = z_2$, $\overrightarrow{OC} = z_3$ and $\arg(z_2 - z_1) = \alpha$:



(i) Find $\arg(z_2 - z_3)$ in terms of α .

1

Let $\beta = \arg(z_2 - z_3)$.

$$\left(\beta + \frac{\pi}{3}\right) + \left(\frac{\pi}{3} - \alpha\right) = \pi \quad (\text{co-interior } \angle\text{s, } \parallel \text{ lines})$$

$$\beta + \frac{2\pi}{3} - \alpha = \pi$$

$$\therefore \beta = \arg(z_2 - z_3) = \alpha + \frac{\pi}{3}$$

1 - correct solution

(ii) Find the modulus and argument of $\frac{z_2 - z_1}{z_3 - z_2}$.

3

Modulus = 1 (since $\triangle$ is equilateral)

In the diagram,

$$\theta + \alpha + \frac{\pi}{3} = \pi \quad (\angle \text{sum of straight } \angle)$$

$$\theta = \frac{2\pi}{3} - \alpha$$

$$\arg(z_3 - z_2) = -\theta = \alpha - \frac{2\pi}{3}$$

$$\therefore \arg\left(\frac{z_2 - z_1}{z_3 - z_2}\right) = \arg(z_2 - z_1) - \arg(z_3 - z_2)$$

(1) - correct modulus

Then

$$= \alpha - \left(\alpha - \frac{2\pi}{3}\right)$$

$$= \frac{2\pi}{3}$$

(1) significant progress for arg.
(2) correct soln

Question 15 (6 marks)

Given $P(z) = z^5 - 1$ and $z \in \mathbb{C}$:

(i) Find the roots of $P(z) = 0$.

2

$$1, \text{cis } \pm \frac{2\pi}{5}, \text{cis } \pm \frac{4\pi}{5}$$

1 - obtain 1 and one other correct root

2 - correct roots (all)

(ii) Express $P(z)$ as a product of real linear and real irreducible quadratic factors.

2

$$P(z) = (z-1)(z^2 - 2z \cos \frac{2\pi}{5} + 1)(z^2 - 2z \cos \frac{4\pi}{5} + 1)$$

1 - obtain $(z-1)$ and two partially correct quad. factors

2 - correct factors

(iii) Given the identity:

$$z^5 - 1 = (z-1)(z^4 + z^3 + z^2 + z + 1)$$

Find the value of:

$$(1 + \cos \frac{2\pi}{5})(1 + \cos \frac{4\pi}{5})$$

2

$$P(z) = z^5 - 1 \quad \text{Let } Q(z) = (z^2 - 2z \cos \frac{2\pi}{5} + 1)(z^2 - 2z \cos \frac{4\pi}{5} + 1)$$

$$= z^4 + z^3 + z^2 + z + 1$$

Method 1

$$\text{Now } Q(-1) = (1 + 2 \cos \frac{2\pi}{5} + 1)(1 + 2 \cos \frac{4\pi}{5} + 1) = 1 - 1 + 1 - 1 + 1$$

$$\therefore 2(1 + \cos \frac{2\pi}{5}) \cdot 2(1 + \cos \frac{4\pi}{5}) = 1$$

(1) substitute $z = -1$

(2) correct soln.

11

$$\therefore (1 + \cos \frac{2\pi}{5})(1 + \cos \frac{4\pi}{5}) = \frac{1}{4}$$

Method 2

$$(z^2 - 2z \cos \frac{2\pi}{5} + 1)(z^2 - 2z \cos \frac{4\pi}{5} + 1) = z^4 + z^3 + z^2 + z + 1 = Q(z)$$

The roots of $Q(z)$ are $\text{cis}(\pm \frac{2\pi}{5})$, $\text{cis}(\pm \frac{4\pi}{5})$

$$\alpha + \beta + \gamma + \delta = -\frac{b}{a} = \text{cis } \frac{2\pi}{5} + \text{cis } -\frac{2\pi}{5} + \text{cis } \frac{4\pi}{5} + \text{cis } -\frac{4\pi}{5}$$

$$-1 = 2 \cos \frac{2\pi}{5} + 2 \cos \frac{4\pi}{5}$$

$$\therefore \cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2} \quad (1)$$

$$\begin{aligned} \Sigma \alpha \beta &= +\frac{c}{a} = \text{cis } \frac{2\pi}{5} \cdot \text{cis } -\frac{2\pi}{5} + \text{cis } \frac{2\pi}{5} \cdot \text{cis } \frac{4\pi}{5} + \text{cis } \frac{2\pi}{5} \cdot \text{cis } -\frac{4\pi}{5} \\ &+ \text{cis } -\frac{2\pi}{5} \cdot \text{cis } \frac{4\pi}{5} + \text{cis } -\frac{2\pi}{5} \cdot \text{cis } -\frac{4\pi}{5} \\ &+ \text{cis } \frac{4\pi}{5} \cdot \text{cis } -\frac{4\pi}{5} \end{aligned}$$

$$1 = 1 + \text{cis } \frac{2\pi}{5} \cdot \text{cis } \frac{4\pi}{5} + \text{cis } \frac{2\pi}{5} \cdot \text{cis } -\frac{4\pi}{5}$$

$$+ \text{cis } -\frac{2\pi}{5} \cdot \text{cis } \frac{4\pi}{5} + \text{cis } -\frac{2\pi}{5} \cdot \text{cis } -\frac{4\pi}{5} + 1$$

$$-1 = \text{cis } \frac{2\pi}{5} (\text{cis } \frac{4\pi}{5} + \text{cis } -\frac{4\pi}{5}) +$$

$$\text{cis } -\frac{2\pi}{5} (\text{cis } \frac{4\pi}{5} + \text{cis } -\frac{4\pi}{5})$$

(1) correct $\Sigma \alpha$.

$$= (\text{cis } \frac{4\pi}{5} + \text{cis } -\frac{4\pi}{5})(\text{cis } \frac{2\pi}{5} + \text{cis } -\frac{2\pi}{5})$$

(2) correct soln.

$$= 2 \cos \frac{4\pi}{5} \cdot 2 \cos \frac{2\pi}{5}$$

$$\therefore \cos \frac{2\pi}{5} \cdot \cos \frac{4\pi}{5} = -\frac{1}{4} \quad (2)$$

$$\text{Now } (1 + \cos \frac{2\pi}{5})(1 + \cos \frac{4\pi}{5}) = 1 + (\cos \frac{4\pi}{5} + \cos \frac{2\pi}{5}) + \cos \frac{2\pi}{5} \cos \frac{4\pi}{5}$$

$$= 1 + (-\frac{1}{2}) + (-\frac{1}{4})$$

$$= \frac{1}{4}$$

Method 3.

$$(z^2 - 2z \cos \frac{2\pi}{5} + 1)(z^2 - 2z \cos \frac{4\pi}{5} + 1) = z^4 + z^3 + z^2 + z + 1$$

Equating coefficients of z :

$$-2 \cos \frac{2\pi}{5} - 2 \cos \frac{4\pi}{5} = 1$$

$$\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$$

Equating coefficients of z^2

$$1 + 1 + 4 \cos \frac{2\pi}{5} \cos \frac{4\pi}{5} = 1$$

$$4 \cos \frac{2\pi}{5} \cos \frac{4\pi}{5} = -1$$

$$\cos \frac{2\pi}{5} \cos \frac{4\pi}{5} = -\frac{1}{4}$$

$$\text{Now } (1 + \cos \frac{2\pi}{5})(1 + \cos \frac{4\pi}{5})$$

$$= 1 + (\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5}) + \cos \frac{2\pi}{5} \cos \frac{4\pi}{5}$$

$$= 1 + -\frac{1}{2} + -\frac{1}{4}$$

$$= \frac{1}{4}$$

(1) correctly equate coefficients

(2) correct solution